



2024 ASABE Student Robotics Challenge **Robot Design Report**

Division : Beginner

Team Name : Earth Online

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CATALOG

1 TEAM MEMBERS	1
2 ESTABLISHMENT OF NEED AND BENEFIT TO AGRICULTURE	2
2.1 THE NECESSITY OF PRUNING UNHEALTHY LEAVES	2
2.2 THE IMPORTANCE OF MECHANIZED REMOVAL OF UNHEALTHY LEAVES..	2
2.3 BENEFITS TO AGRICULTURE	3
3 DESIGN OBJECTIVES	4
4 ROBOT DESIGN	4
4.1 HARDWARE	4
4.2 SOFTWARE	4
4.3 ULTRASONIC AND INFRARED POSITIONING SYSTEM	5
4.4 HOOK TYPE	5
5 PARTS LIST AND TABLE	5
6 HARDWARE DESCRIPTION	6
6.1 OPERATION MODULE	6
6.1.1 PICKING DEVICE	6
6.2 MOTION MODULE	6
6.2.1 HARVESTING MOTION	6
6.3 THE ITERATIVE PROCESS OF MECHANICAL DESIGN	6
6.3.1 THE FIRST VERSION--ORDINARY HOOK	7
6.3.2 THE SECOND VERSION--BITE HOOKS	7
6.4 CIRCUIT & ELECTRONIC COMPONENT MODULE	7
6.4.1 OPENMV	7
6.4.2 ARDUINO	8
6.4.3 PRINTED CIRCUIT BOARD	9
7 SOFTWARE OVERVIEW	9
7.1 OVERALL WORKFLOW	9
7.2 POSITIONING MODULE	11
7.3 ACQUISITION MODULE	12

1 Team Members

In preparation for the 2024 ASABE Student Robotics Challenge, five of us formed a team which consisting of four freshmen and one junior, each with their own strengths and efforts. We won first prize in the 2023 China Agricultural Robotics Competition, out of only five teams in the country winning the award. The details of the members are listed below.

Table 1-1 Profiles of team members

Name	Specialties	Assigned tasks	Identity
Jiahao Wang	·SolidWorks 3D Modeling	• Motion Module Design	Freshman
	·SolidWorks Model Rendering	• Operation Module Design	
	·Mechanical Design and Assembly	• Assembly and Rendering	Captain
	·Programming and Debugging	• Report Writing	
	·Copywriting	• Bookkeeping and Reimbursement	
Boyue Qiu	·PCB Design	• Circuit Module Design	Freshman
	·Hardware Welding	• Arduino Software Design	ASABE
	Circuit wiring	• Tracking Module Design	Member
Jiakang Jin	·Sensor Applications	• Tracking Module Design	Freshman
	·Programming and Debugging	• Arduino Software Design	ASABE
	·Computer Vision	•Hardware Processing	Member
	·Embedded System Design		
	·Computer Vision	• Identification Module Design	
Huachen Wei	·Embedded System Design	• Operation Module Design	Freshman
	·Programming and Debugging	• Tracking Module Design	ASABE
	·SolidWorks 3D Modeling	• Arduino Software Design	Member
	·Computer Vision		
Lingsong Xu	• Operation Module Design		Junior
	·Embedded System Design·Sensor Applications		ASABE
			Member

2 Establishment of Need and Benefit to Agriculture

2.1 The necessity of pruning unhealthy leaves

Pruning unhealthy leaves is essential for several reasons:

Disease Control: Removing infected leaves helps contain the spread of disease to other parts of the plant and to nearby plants, thereby limiting the impact of the disease.

Pest Prevention: unhealthy leaves can attract pests, which can further damage the plant or spread to other healthy plants. Pruning helps mitigate this risk.

Improved Air Circulation: Pruning enhances airflow around the plant, reducing humidity levels and thereby decreasing the likelihood of fungal and bacterial infections.

Resource Allocation: By removing unhealthy parts, the plant can allocate its energy and nutrients towards healthier growth and the production of new foliage or flowers.

Aesthetic Appeal: Regular pruning keeps plants looking healthy and well-maintained, enhancing the overall appearance of your garden or indoor plant collection.

In summary, pruning unhealthy leaves is a proactive measure that helps ensure the health, growth, and aesthetics of plants.

2.2 The importance of mechanized removal of unhealthy leaves

The mechanized removal of unhealthy leaves brings several advantages over traditional manual methods, particularly in larger agricultural operations or settings where efficiency and precision are critical. Here are some key points highlighting the importance of mechanized removal:

Efficiency and Speed: Machines can remove unhealthy leaves faster than manual labor, significantly improving the speed of the process. This is particularly valuable in large-scale agricultural operations where time is of the essence.

Consistency and Precision: Mechanized systems can be programmed to identify and remove only the unhealthy parts of a plant with high accuracy. This consistency ensures that healthy parts of the plant remain intact, optimizing overall plant health.

Labor Savings: The use of machines reduces the reliance on manual labor, which can be both time-consuming and costly. This not only lowers labor costs but also addresses shortages in manual labor availability.

Minimized Human Error: Human workers may miss some unhealthy leaves or inadvertently damage healthy parts, whereas properly calibrated machines can operate with consistent precision, minimizing errors.

Scalability: Mechanized systems can be scaled up to handle large volumes of plants, making them suitable for commercial agricultural enterprises and large greenhouses. This scalability is less feasible with manual pruning.

Health and Safety: Reducing the need for manual labor in potentially hazardous environments (e.g., exposure to chemical treatments or working in difficult weather conditions) enhances worker safety and reduces health risks.

Cost-Effectiveness: Over time, the investment in mechanized systems can be more cost-effective than ongoing manual labor costs, particularly for large operations. The initial investment in machines can be offset by increased productivity and reduced labor expenses.

Integration with Technology: Modern mechanized systems can integrate with other agricultural technologies, such as sensors, data analytics, and remote monitoring systems, to further optimize plant health and yield.

In summary, the mechanized removal of unhealthy leaves offers substantial benefits in terms of efficiency, precision, evenness, and scalability, making it a valuable approach in modern agriculture.

2.3 Benefits to Agriculture

The mechanized removal of unhealthy leaves offers numerous benefits to agriculture, enhancing productivity, efficiency, and sustainability. Here are some detailed benefits:

Improved Crop Health: By promptly removing unhealthy leaves, mechanized systems help prevent the spread of diseases to healthy parts of the plant and neighboring plants. This contributes to overall better plant health and higher yields.

Increased Yield and Quality: Healthier plants with fewer unhealthy leaves typically produce higher-quality produce. This leads to better market value and potentially higher revenue for farmers.

Optimized Use of Resources: Mechanized removal ensures that only the affected parts are removed, preserving more of the healthy plant. This optimization can result in better utilization of water, nutrients, and other resources, as the remaining healthy plant can continue growing efficiently.

Reduced Pesticide Use: By controlling disease spread more effectively through physical removal, there may be less need for chemical interventions such as pesticides and fungicides. This can lead to cost savings and environmental benefits by reducing chemical runoff and residue on produce.

Labor Efficiency: Mechanized systems reduce the need for extensive manual labor, freeing up workers for other essential tasks and reducing overall labor costs. This is especially valuable in regions facing labor shortages.

Scalability and Adaptability: Mechanization can be scaled to match the needs of farms of various sizes, from small operations to large commercial farms. Advanced systems can be adapted for different crops and growing conditions, making them versatile tools in modern agriculture.

Data Integration and Precision Agriculture: Modern mechanized systems can be integrated with digital agriculture technologies, such as sensors, data analytics, and GPS. This integration allows for more precise monitoring and management of crop health, leading to better decision-making and optimized farming practices.

Consistency and Reliability: Machines can operate consistently without fatigue, ensuring that the removal process is thorough and uniformly applied across the entire field. This reliability reduces the risk of leaving unhealthy leaves untreated.

Environmental Sustainability: By minimizing the use of chemicals and improving resource utilization, mechanized removal contributes to more sustainable farming practices. Healthier crops and reduced pesticide use align with environmentally friendly agricultural practices.

Economic Benefits: The long-term economic benefits include reduced labor costs, higher crop yields, and potentially higher profits from better quality produce. Moreover, the investment in mechanization can be recouped over time through these efficiencies.

In conclusion, the mechanized removal of unhealthy leaves offers substantial benefits to agriculture, contributing to healthier crops, increased productivity, and more sustainable farming practices. It represents a critical advancement in modern agricultural techniques, helping farmers meet the challenges of feeding a growing global population while sustaining the environment.

3 Design Objectives

The robot designed for this competition should have the ability to identify healthy leaves, flowers and unhealthy leaves, cut unhealthy leaves to retain healthy leaves and flowers, and complete the counting display. In order to achieve the goal perfectly, a detailed and specific division is made to complete the task efficiently and accurately. The breakdown is as follows:

- (1) The robot needs to have a navigation function and no guide is required.
- (2) Accurate identification of different leaves and flowers in the competition, and the recognition results are generated into sev files and stored in USB drives.
- (3) Cut the unhealthy leaves to prevent the healthy leaves from falling.
- (4) The robot should work independently without any external communication or human intervention.
- (5) Trace without a guide and ridge, despite a guide.
- (6) The machine can count automatically.
- (7) No human intervention to the robot.
- (8) Max robot size of 12"× 12" ×12" at the start of each time trial.

4 Robot design

4.1 Hardware

We have broken down the overall objective into five design modules: the operation module, the motion module, the circuit & electronic components module, the tracking module, and the identification module. This modular design approach allows us to work with clearer objectives and facilitates the division of labor, management and schedule advancement.

Therefore, we use eight hooks and copper pillars, using lifting devices and optional devices to align the operating module with the blades, and then the corresponding hooks will be driven by the servo to rotate and engage with the copper pillars, and then the lifting device lifts up to pull out the blades.

4.2 Software

The prerequisite for automating all leaf cutting is successful tracking. In the tracking module, we use OpenMV to align with the plant's ridge and the plant's ridge to determine the robot's position, and use Arduino to control the stepper motor to correct the robot's position offset. In the

recognition module, we use OpenMV to identify plant stems and cameras to determine the position of individual leaves. After the identification information is transmitted to the Arduino, the Arduino will control the operating module and the motion module for the harvesting operation.

4.3 Ultrasonic and Infrared Positioning System

The robot stops at the desired position via OpenMV, determines the distance by the given number of degrees of the two telescopic servos, and adjusts the parallelism between the robot and the field board to achieve more precise operation.

4.4 Hook type

The Removal is rotated out with a hook-style structure to reduce damage to healthy blades and improve efficiency. In addition, unhealthy leaves can be easily separated by sliding upwards, allowing for efficient and simple removal. In addition, since the telescopic boom is long enough to reach both sides easily, a bilateral structure is used to improve harvesting efficiency.

5 Parts List and Table

The total cost of our robot was \$506.55, as shown in Table 1.

Table 1 Parts List and Cost

No.	Item	Type (Spec)	Qty	Unit Price (RMB)	Total Price (RMB)
1	Arduino	Mega2560 Pro	2	68	136
2	OpenMV	OpenMV4 CAM H7Plus	2	349	698
3	42 Stepper Motor	42BYGH47	4	69	276
4	Stepper Motor Frame	42 Stepper Motor	4	2.04	8.16
5	42 Stepper Motor Driver	/	4	26.9	107.6
6	Servo Motor	SG90 MG90s	16	10.69	171.04
7	Servo	32 kg	2	286.2	572.4
8	Printed Circuit Board	Customized	2	11.8	23.6
9	Carbon Fiber Plate	5 mm/Customized	1	359	359
10	Acrylic Plate	5 mm/Customized	1	79	79
11	Copper Pillar	40 mm/60 mm/65 mm/70 mm/90 mm	/	/	15
12	Screw	4 mm*10/4 mm*15	/	/	8
13	Nuts	4 mm/3 mm	/	/	6
14	4S Battery	4S 16.8 V	1	49.9	49.9
15	3D Printing Material	Photosensitive Resin	2	78	156
16	Angle Iron	Type 20	12	1.5	18
17	Wires	/	/	/	10
18	Aluminum profiles	TD2020	/	/	253.2
19	28 linear actuator motor guides	T6*1-200 mm	2	110	220
20	Mecanum Wheel	60 mm	4	130	520
Sum			3681.3¥ /506.55\$		

6 Hardware Description

In order to meet the requirements of the competition task, we have designed picking device, separation device, collection device, telescopic device.

6.1 Operation Module

6.1.1 Picking Device

Unhealthy leaves are removed using a hook-like structure with copper pillars and hooks at the ends. The unhealthy leaf is bitten off by the bite of the copper pillar and the hook, so that the leaf gap is smaller than the leaf size. Positioning is done by a rotating turntable, which is driven by a screw slide and a servo.

6.2 Motion Module

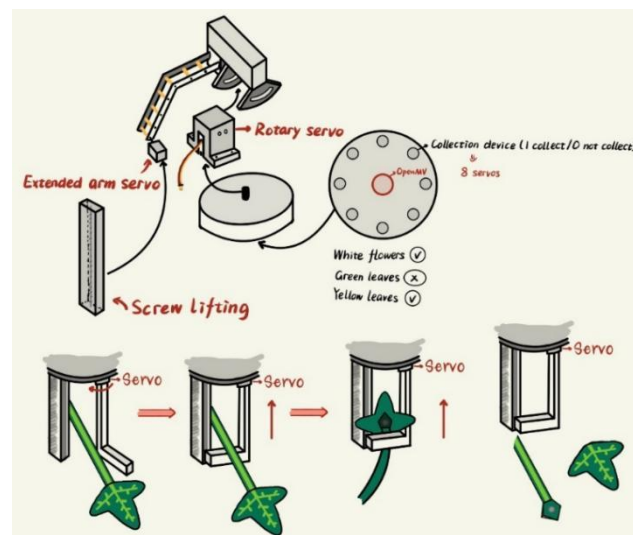


Figure 1 Harvesting Motion

6.2.1 Harvesting Motion

The movement of the leaf shearing mechanism includes extension, rotation and lifting. We use a robotic arm mechanism driven by a servo motor to realize the expansion and contraction of the shearing mechanism. In order to accurately control the position of the picking mechanism and ensure a large adjustable range, we chose a 32 kg motor servo with a 180° fully controllable angle. For the rotational motion of the positioning mechanism, we use a 360° servo servo to rotate. At the same time, the screw slide table is used to rotate the lifting part of the whole mechanism, and a innovative suspension mechanism is used to ensure that the mechanism can be cut parallel to the table surface, as shown in Figure 1.

6.3 The Iterative Process of Mechanical Design

We have been preparing for the ASABE Student Robotics Challenge since February. During this process, we also completed several mechanical designs. Although we do not need to provide our mechanical design iteration process, we are still presenting them in the hope that we can communicate more widely with students from all over the world.

6.3.1 The first version--Ordinary hook

At first, it was only hooked by an ordinary hook (as shown in Figure 2), but in practice, we found that such a single hook could easily fail to get the blade off, and the error tolerance rate was low.

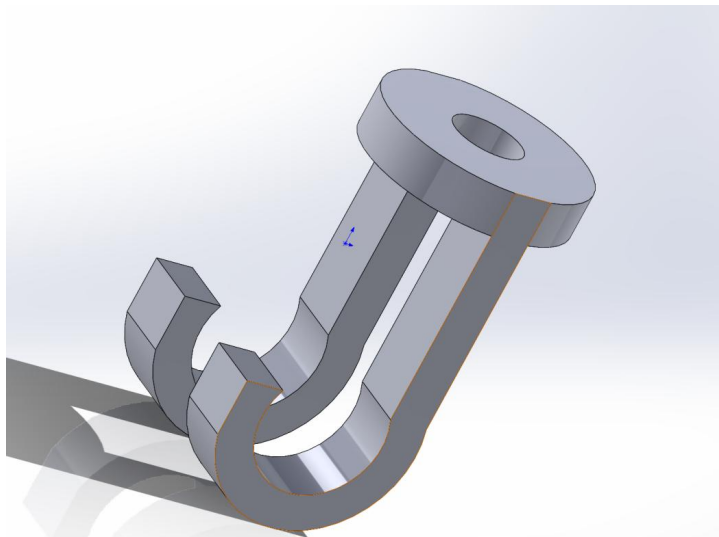


Figure 2 The first version of hook

6.3.2 The second version--Bite hook

The occlusion of the copper pillar and hook increases the fault tolerance and also forms a closed structure to prevent the blades from popping out, as shown in Figure 3.

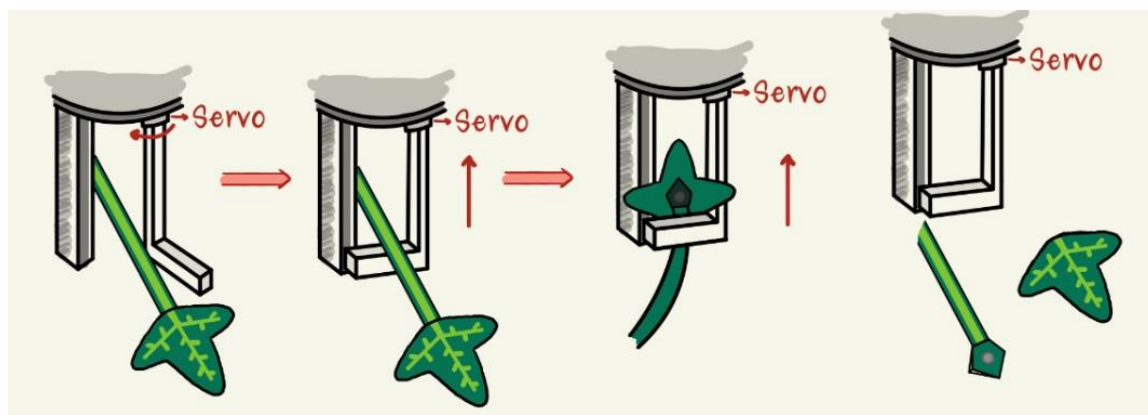


Figure 3 The second version of hook

6.4 Circuit & Electronic Component Module

6.4.1 OpenMV

OpenMV is a compact computer module designed for AI and robotics applications. It is

powered by an NVIDIA Maxwell GPU with 128 CUDA cores and a quad-core ARM Cortex-A57 CPU, making it capable of performing complex AI computations and running multiple applications simultaneously.

The OpenMV module features 4 GB of LPDDR4 memory, a microSD slot for additional storage, Gigabit Ethernet, and support for multiple USB 3.0 and USB 2.0 ports. It also includes a dedicated MIPI-CSI camera connector and HDMI display output for video capture and display capabilities.

This highly versatile module is ideal for a wide range of AI and robotics applications, including object detection, image classification, speech recognition, and autonomous navigation. It can be used in a variety of devices, from small robots to intelligent cameras and drones.

The OpenMV (As shown in Figure 4) has such excellent performance that we use it as our image processing module for picking unhealthy and healthy leaves.



Figure 4 OpenMV

6.4.2 Arduino

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It consists of a programmable microcontroller board, which can be used to create a wide range of interactive objects and projects.

Arduino boards, as shown in Figure 5, are programmed using the Arduino Integrated Development Environment (IDE), which is a user-friendly software application that allows users to write and upload code to the board. The IDE includes a range of libraries and examples that simplify the process of programming and enable users to quickly get started with their projects.

We use Arduino as the main controller to collect data information from various sensors and control various actuators. Arduino also communicates with OpenMV to obtain image data information processed by OpenMV.

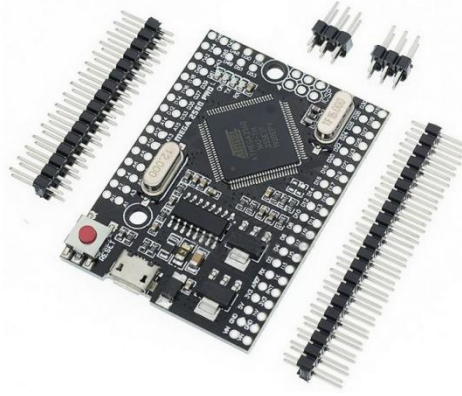


Figure 5 Arduino control board

6.4.3 Printed Circuit Board

We have developed a custom printed circuit board (PCB) that can connect a microcontroller and other electronic components, as shown in Figure 6. This PCB is versatile and has a wide range of interfaces, making it suitable for use in various types of robots.

The PCB includes a Mega2560 Pro intelligent control board. To power the electronic development board and other external sensors, we have included a 5 V voltage regulator module. Additionally, two DC adjustable voltage regulator modules are included, which can adjust the voltage to 6 V and 12 V respectively. These voltage regulators can supply power to external devices such as servos and motors, providing the necessary power for their operation.

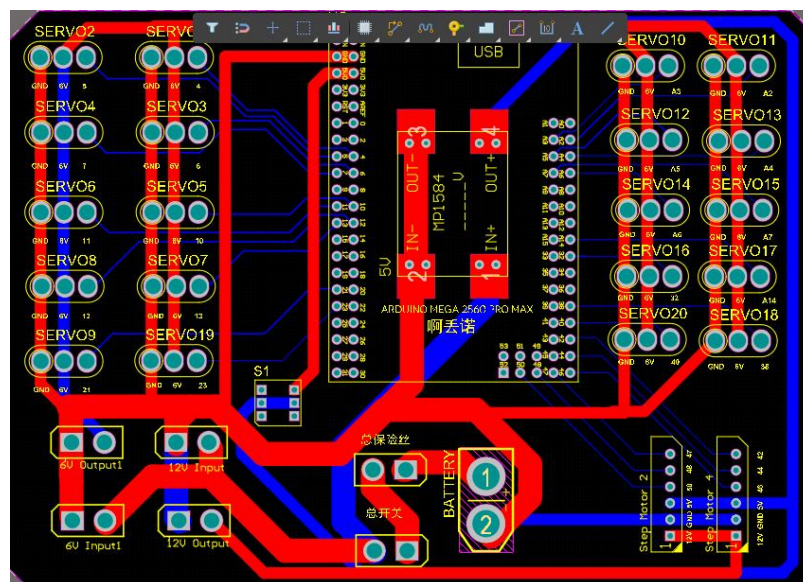


Figure 6 PCB circuit board

7 Software Overview

7.1 Overall Workflow

The robot begins at position A, moves to the midway point of AB (as shown in Figure 7) for initialization, including extending the robotic arm to its initial position, advancing a set distance, and initiating OpenMV parameter transmission. As the robot starts, OpenMV identifies the position of the stem. If it is detected at the designated location in the recognition image, we consider it has reached the midpoint of A1B1. The cutting mechanism maintains a clearance from the stem, allowing the robotic arm to rotate and extend as necessary. Initially, the robot's alignment is checked by assessing whether the arm extension servo has reached its threshold to determine if there is any deviation. If no deviation is detected, only the robot's steering angle is adjusted. Subsequently, OpenMV identifies flowers and leaves to decide whether to proceed with the harvest. Once the harvesting operation is complete, OpenMV recognition is disabled, and the robot moves a set distance before re-enabling it to prevent backtracking. After repeating the cycle 12 times, the robot returns to the end point and stops, completing the race.

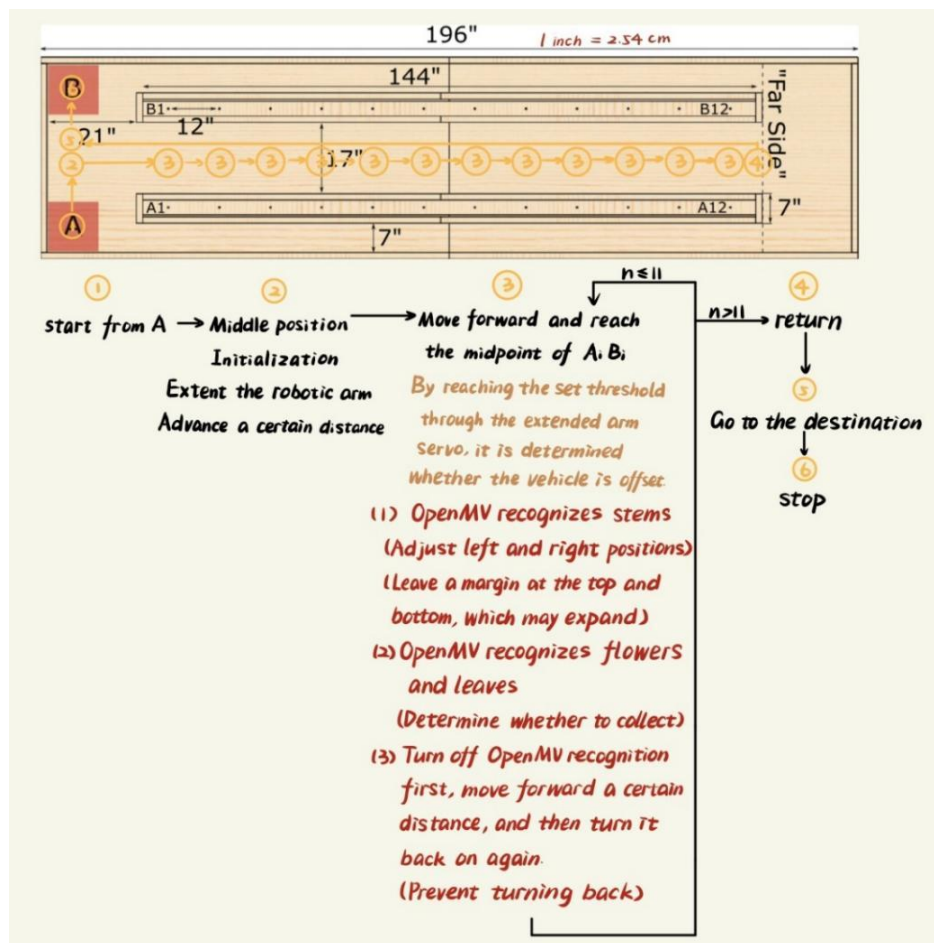


Figure 7 Overall workflow of the robot

7.2 Positioning Module

This competition features only two ridges and a straight track. Our team opted for inter-ridge operation, which means our robot cannot maintain a straight path as effectively as teams operating across ridges. This suggests that our team's positioning module will outperform that of other groups. At the start of the competition, our robot will initially travel a short distance blindly toward the midpoint of AB. Thanks to Mecanum Wheel, directional changes do not require the robot to be rotated, significantly enhancing the stability of its travel path. Subsequently, as the robot advances towards the ridges, it employs OpenMV to identify whether the stem is present at the designated position in the image, ensuring it reaches each specified location during inter-ridge operations. Then, using PID control, adjustments are made to the arm extension servo and lifting motor parameters to precisely center the stem within the acquisition device, as shown in Figure 8.

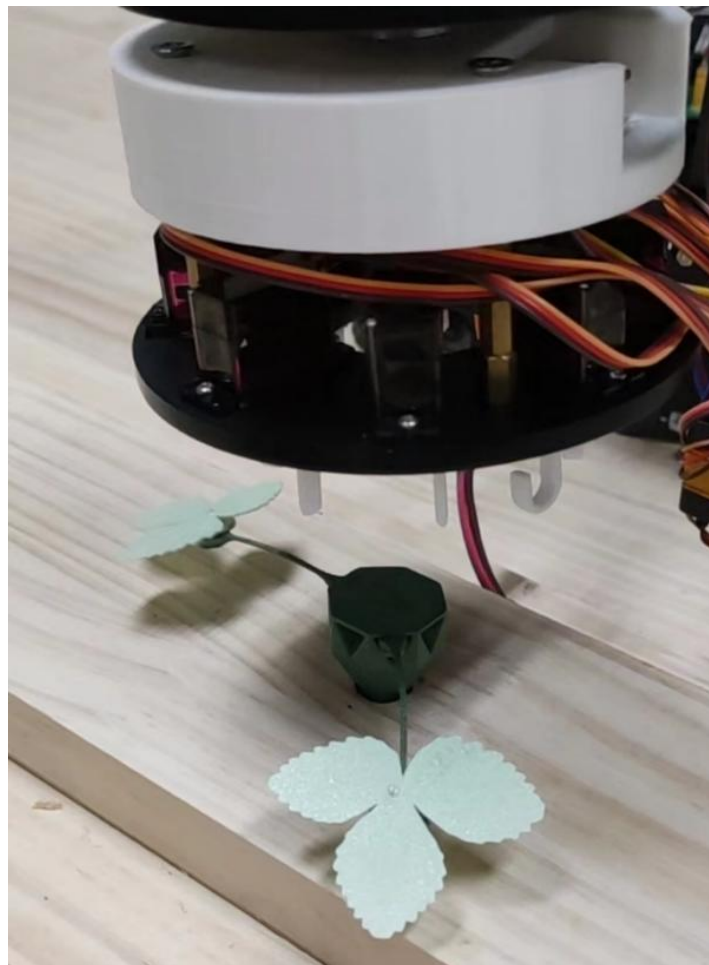


Figure 8 Positioning module

7.3 Acquisition Module

When the robot reaches the designated position of A1B1, it enters the collection state, and the lifting motor and telescopic servo operate simultaneously, causing the disc rotating 8 collection devices to descend to the same height as the collection devices and the branches of flowers and leaves. At this point, OpenMV will detect the recognized image partition blocks and return the corresponding high and low signals to the corresponding interfaces on Arduino in order. It will then control whether the corresponding acquisition device will collect them. If it returns a high level, it will collect them, and if it returns a low level, it doesn't, as shown in Figure 9 . After the collection operation is completed, the lifting motor controls the end effector to rise until the collection device is above a certain position on the stem. Due to the repetitive loop operation, there is no need to return the device to the initialization position, which greatly saves time.

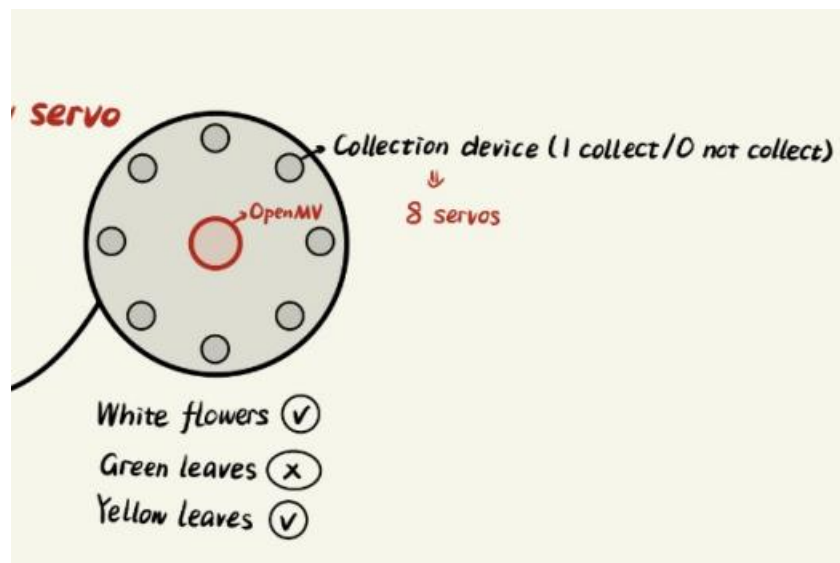


Figure 9 Acquisition Module